

Summary

This chapter explored several algorithms that make use of tree data structures and backtracking, which are features of a problem that is suitable for solving with recursive algorithms. We covered tree data structures, which are composed of nodes that contain data and edges that relate nodes together in parent–child relationships. In particular, we examined a specific kind of tree called a directed acyclic graph (DAG) that is often used in recursive algorithms. A recursive function call is analogous to traversing to a child node in a tree, while returning from a recursive function call is analogous to backtracking to a previous parent node.

While recursion is overused for simple programming problems, it is well matched for problems that involve tree-like structures and backtracking. Using these ideas of tree-like structures, we wrote several algorithms for traversing, searching, and determining the depth of tree structures. We also showed that a simply connected maze has a tree-like structure, and employed recursion and backtracking to solve a maze.